Haysun Hannah Choi

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Education

Yale University Sep 2021 – current
PhD Candidate in Ecology & Evolutionary Biology (E&EB)
Advisor: Richard O. Prum

University of Utah (U of U) Sep 2013 – May 2018
Honors BS in Applied Mathematics, minor in Chemistry; GPA: 3.8/4.0

Scholarships and Grants

Dissertation Improvement Grant Yale Institute for Biospheric Studies ($5000) 2024
Public Humanities Grant Yale Environmental Humanities ($2500) 2023 – 2024
Collections Engagement Grant The Utah Museum of Fine Arts ($4400) 2020
NSF Research Experience for Undergraduates University of Utah Mathematics ($1500) Fall 2017
NSF Research Experience for Undergraduates University of Utah Mathematics ($1500) Spring 2018
Outdoor Education Continuing Learning Grant University of Utah Outdoor Program ($1000) Spring 2017
Honors at Entrance University of Utah Admissions (Tuition) 2013 – 2017
ACCESS Women in Science University of Utah College of Science ($5000) Summer 2013
Regents' Scholarship Utah System of Higher Education ($4000) 2013 –2014

Publications

Choi, H.H. and Prum, R.O. Sexual Selection in the Price Equation. *The American Naturalist.* [In review]
Choi, H.H. and Prum, R.O. A Quantitative, Darwinian Framework for Alternative Mechanisms of Intersexual Selection and Sexual Conflict over Mating. [In prep.]

Presentations and Posters*

Hyperuniformity in Non-Iridescent, Avian Photonic Glasses* June 2026
Choi, H.H., Prum, R.O., and Saranathan V. Bioinspired Materials GRC, Les Diablerets, CH

A Darwinian Framework for Intersexual Mechanisms of Evolution over Mating Feb 2026
Choi, H.H. and Prum, R.O. Edwards Lab, Harvard University, Boston, US

Modeling the Development of Structural Color: Applications of Cahn-Hillard and Persistent Homology May 2025
Choi, H.H., Prum, R.O., Ma, T., and Pedersen M. C. Active Matter Lab, Niels Bohr Institute, Copenhagen, DK

Reproduction in the Price Equation Dec 2024
Choi, H.H. and Prum, R.O. E&EB Graduate Research Symposium, Yale, New Haven, US

A Unified Framework for Mechanisms of Mate Choice, Sexual Conflict, and Intrasexual Competition* July 2024
Choi, H.H. and Prum, R.O. Society for the Study of Evolution Meeting, Montreal, CA

Sexual Selection in the Price Equation* May 2024
Choi, H.H. and Prum, R.O. Sexual Selection in a Changing World Workshop, Erice, IT

Eclectic Forms Dec 2021
Choi, H.H. Environmental Humanities Symposium, Yale, New Haven, US

Form-ing Curiosity Feb 2020
Choi, H.H., Liechty, S. S., Rose, C., and Mershin, A. Center for Bits & Atoms, MIT, Boston, US

A Mathematical Approach to Human Experience May 2018
Choi, H.H., Lawley, S., and Seger, J. Honors College Symposium, University of Utah, Salt Lake City, US

Optimal Predator Search Strategies Using Olfaction and Mean First Passage Time May 2018
Choi, H.H. and Lawley, S. Senior Research Symposium, University of Utah, Salt Lake City, US

External Academic Training

Visiting Researcher Biophysics & Biocomplexity Group, Niels Bohr Institute	Spring 2025
Participant Sexual Selection in a Changing World Workshop, Università degli Studi di Padova	Summer 2024
Participant Modeling & Theory in Population Biology Workshop, Banff International Research Station	Spring 2024
Student Evolutionary Quantitative Genetics Workshop, University of Washington	Summer 2022
Coordinator Multivariate Evolutionary Theory Reading Group, University of Utah	Summer 2022
Student Network Science & Complexity Reading Group, Northeastern University	Spring 2022

Community Leadership and Outreach

Graduate Mentor Yale E&EB	2023 – 2024
Hutchinson Speaker Coordinator Yale E&EB	Fall 2024
Elected Graduate Student Representative Yale E&EB	2022 – 2023
DEI Committee Member Yale E&EB	2022 – 2023
Conference Delegate Biology at Columbia, Rutgers, Princeton, UPenn, Yale E&EB	Spring 2023
Graduate Professional Fellow Yale Center for British Art	Spring 2023
Reader The Yale Review	2021 – 2022
Assistant Editor University of Utah Special Collections	2021
Equipment Studio Monitor University of Utah Book Arts Program	2017 – 2019
Grader University of Utah Mathematics	2018 – 2019
Backcountry Trip Leader; Trainer University of Utah Outdoor Program	2014 – 2018
D.C. National Scholar University of Utah Hinckley Institute of Politics	Summer 2016

Collaborations and Creative Projects

Workshop, Consequences of Environmental Stochasticity for Spatial Dynamics National Science Foundation, Yale Fine Arts, Yale E&EB, Princeton E&EB	Spring 2022
Journal; Short Film, Abandon Shape; From the Marrow University of Utah Honors College, University of Utah Special Collections, Utah Museum of Fine Arts	Fall 2021
Artists Book, Translational Reconciliation University of Utah Book Arts Program	Summer 2021
Workshop, Honing Curiosity MIT Center for Bits & Atoms	Winter 2020
Undergraduate Thesis, A Mathematical Approach to Human Experience University of Utah Honors College	May 2018

Teaching Fellowships & Assistantships

Ornithology Lecture and Lab, Yale E&EB	Spring 2023
Introduction to Evolution and Ecology, Yale E&EB	Spring 2022, Fall 2024
Terrestrial Arthropods Lecture and Lab, Yale E&EB	Fall 2021
Biology of Variation, University of Utah Biology	Fall 2019
Rock Climbing I-II, University of Utah Parks, Recreation, & Tourism	Fall 2016

Professional Experience

Sous Chef Nohm, Salt Lake City, US	2019 – 2021
Logistics Coordinator; Assistant Instructor Northwest Outward Bound School, Mazama, US	2018 – 2019
Medical Clerk; First Responder Alta Ski Resort, Sandy, US	2016 – 2018